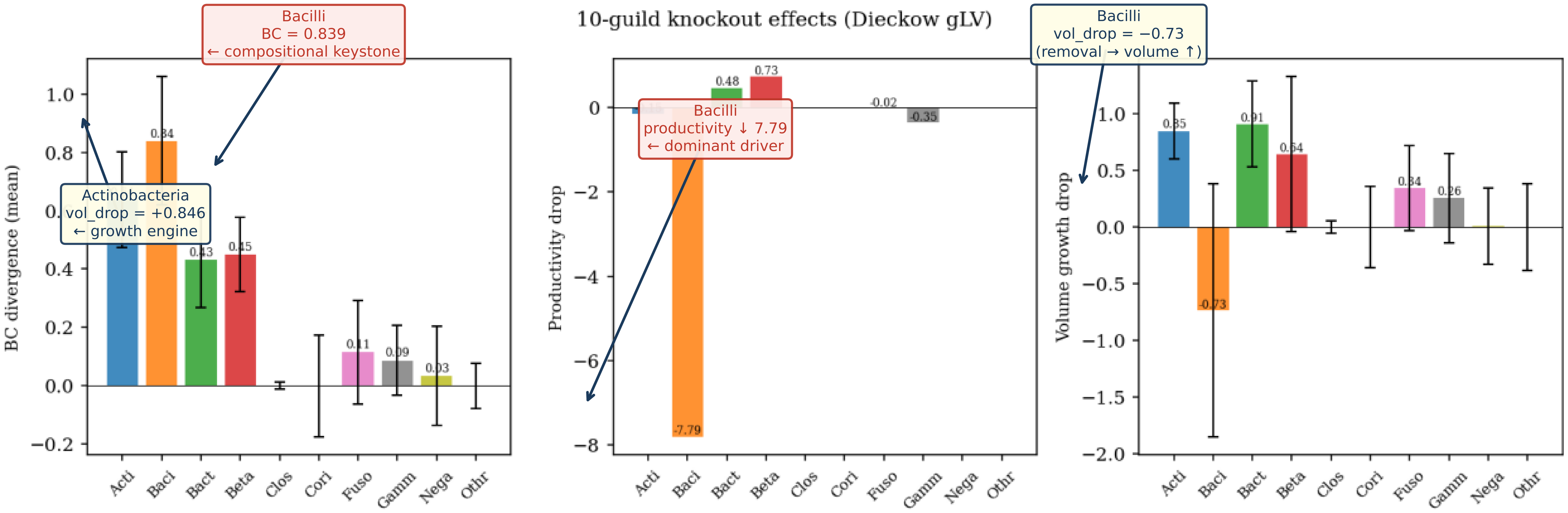
Guild-Level Dynamical Analysis

Keystone · Network Centrality · Relapse Dynamics · Prevention · Phase Diagram

Dieckow 2024 · 10-guild gLV · N = 10 patients · Nishioka (2026)

Guild-Level Keystone Analysis: Knockout BC Divergence

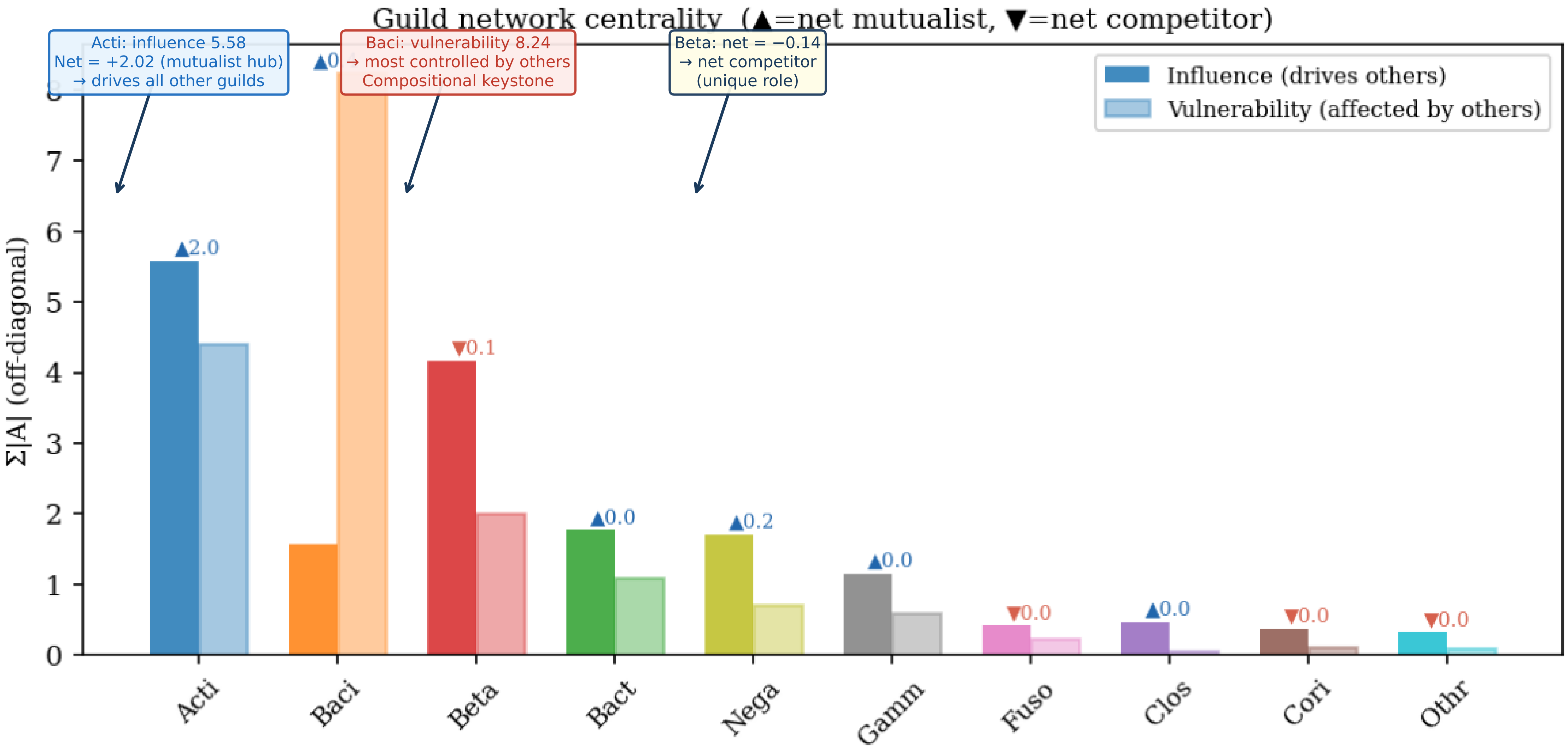
Remove one guild → simulate gLV 21 d → measure BC shift, productivity drop, volume change



Bacilli: highest BC divergence (0.839) — compositional keystone. Removal increases total volume → competitive suppressor. | Actinobacteria: vol_drop = +0.846 → growth engine. Strongest single interaction: Acti → Baci A = +3.43.

Guild Network Centrality: Influence, Vulnerability, Net Role

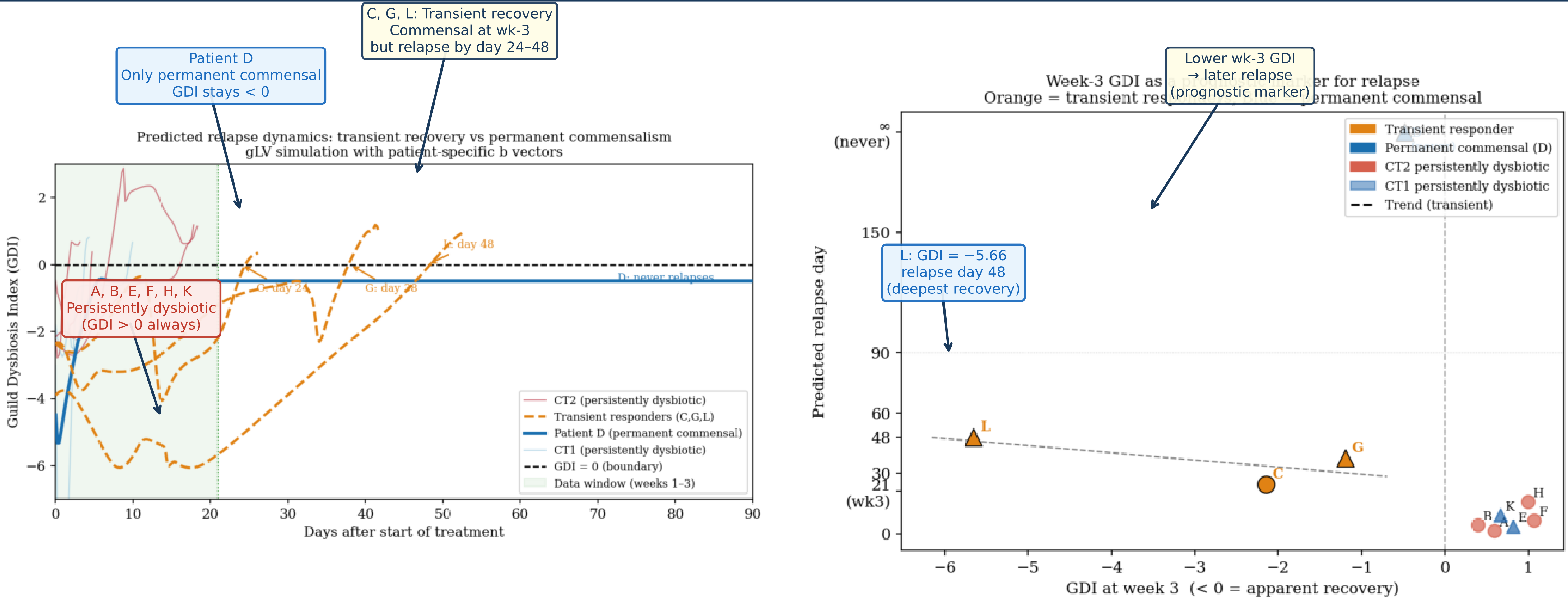
Influence = Σ|A_ji| (how much guild i drives others) | Vulnerability = Σ|A_ij| (how much others drive guild i)



Actinobacteria: influence = 5.58 (highest), net = +2.02 → mutualist hub. Bacilli: vulnerability = 8.24 → most regulated by others. Betaproteobacteria: net = −0.14 → net competitor. A matrix encodes dysbiotic attractor via Acti–Baci–Beta triad.

Predicted Relapse Dynamics: Three Patient Archetypes

gLV simulation t = 150 d with patient-specific b vectors | GDI = log(φ_dys) − log(φ_com)



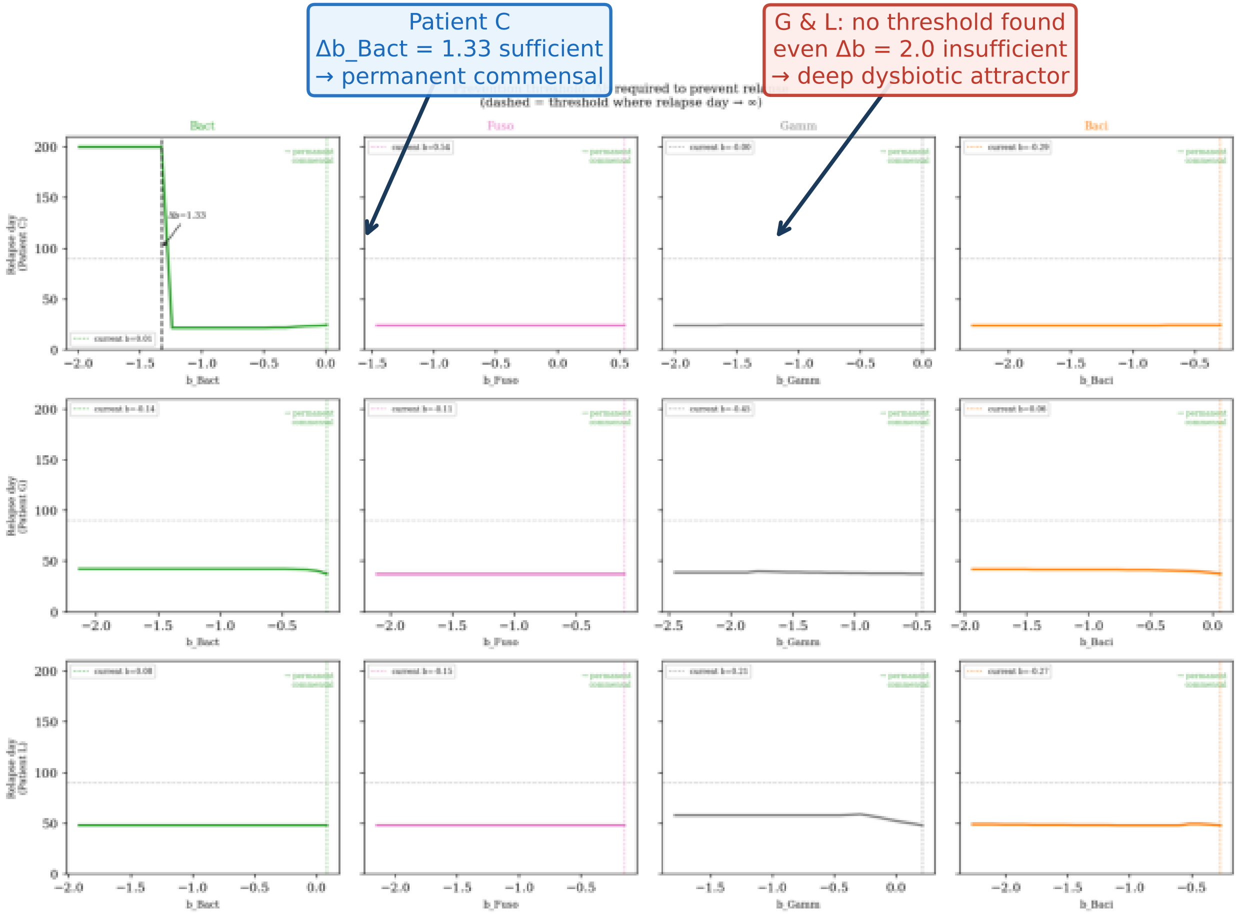
(a) GDI trajectories to 90 d

(b) Week-3 GDI as prognostic marker

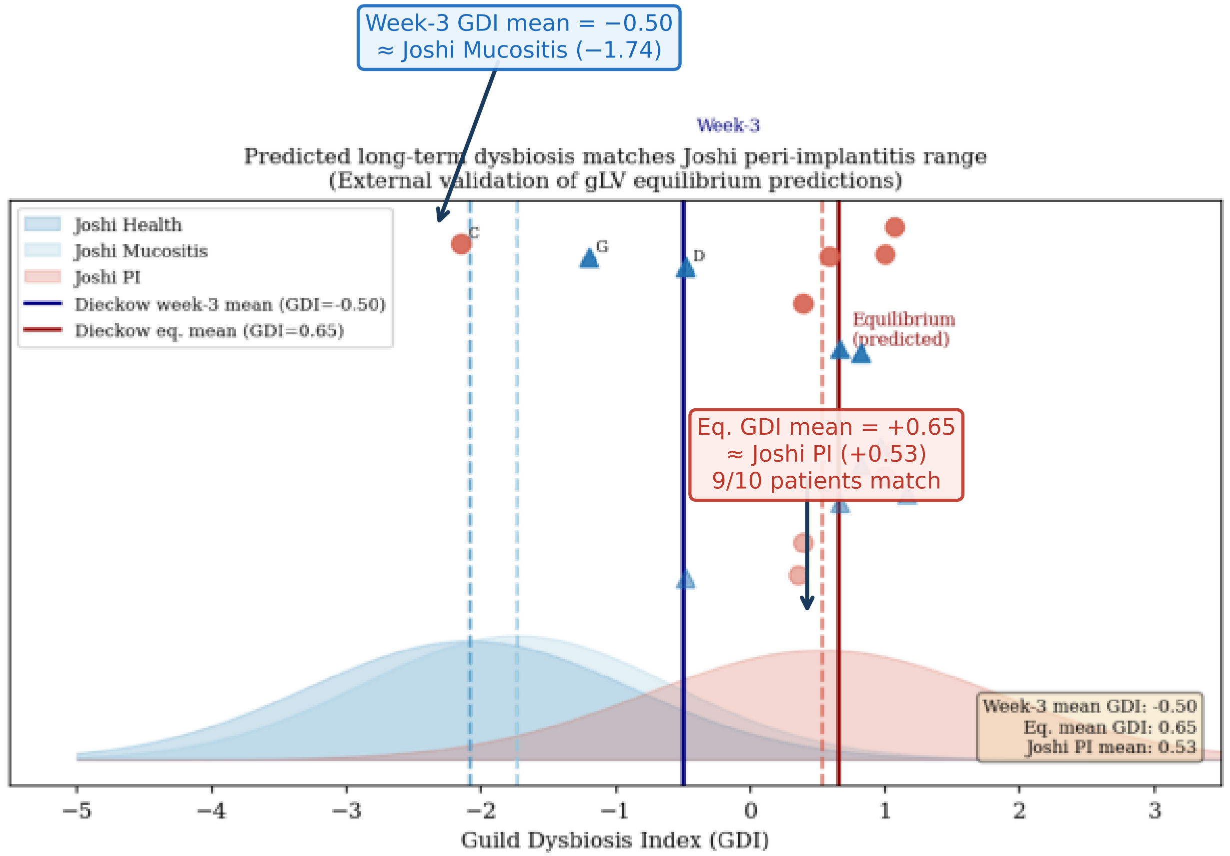
① Persistent dysbiotic (A,B,E,F,H,K): GDI > 0 throughout. ② Transient responders (C→24d, G→38d, L→48d): commensal at wk-3, dysbiotic by day 50. ③ Permanent commensal (D only): stable GDI < 0. → Week-3 GDI as prognostic marker: lower = longer remission.

Prevention Threshold & External Validation (Joshi 2025)

Min. Δb_i to prevent relapse (single-guild sweep) | Model eq. GDI vs peri-implantitis reference distribution



(a) Required Δb per guild for prevention

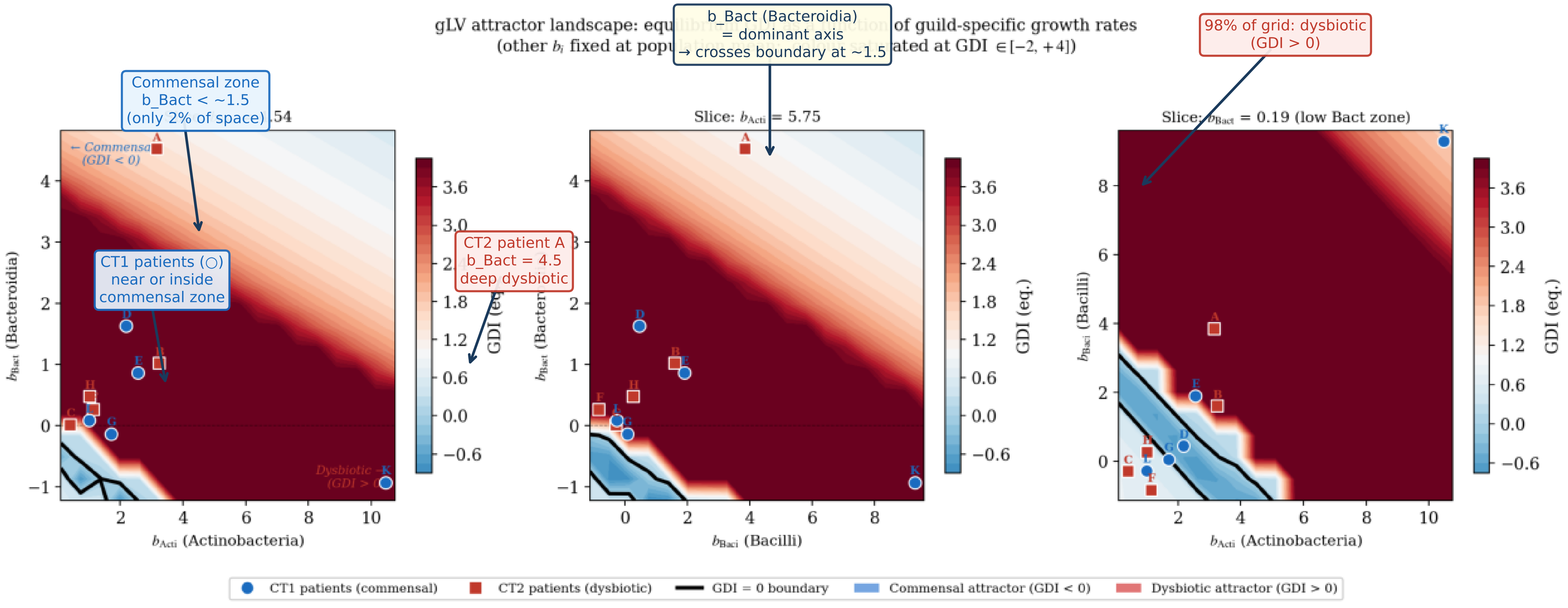


(b) External validation: Joshi 2025 peri-implant cohort

Prevention: only Patient C convertible by single-guild intervention (Δb_Bact = 1.33). G & L require multi-guild change → deep attractor. | Joshi validation: predicted eq. GDI = +0.65 ≈ Joshi PI mean +0.53 (9/10 patients; difference < 0.12 < Joshi SD = 1.30).

3D Attractor Landscape: GDI = 0 Boundary in Growth-Rate Space

18^3 grid, Acti × Baci × Bact, t = 80 d | Blue = commensal | Red = dysbiotic | Black = GDI = 0 boundary



98% of parameter space is dysbiotic (GDI > 0). Commensal region confined to b_Bact < ~1.5 — Bacteroidia growth rate is the dominant control variable. CT1 patients with low b_Bact cluster near boundary; CT2 patient A (b_Bact = 4.5) is deep in dysbiotic zone. → Structural attractor robustness — not initial-condition sensitive.